

Mini-Project Proposal

Title

Adaptive Hybrid CRC–Hamming Error Control System for Reliable Data Transmission

Team Details

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Abstract

Reliable data communication is critical in modern systems such as wireless networks, IoT devices, embedded systems, and digital memory architectures. During transmission, data can be corrupted due to noise, interference, signal attenuation, or hardware faults. Even a single bit error can compromise the integrity of an entire message. To address this, error-control techniques are implemented at the Data Link Layer to detect and correct transmission errors. Cyclic Redundancy Check (CRC) is widely used for detecting both random and burst errors due to its strong polynomial-based detection capability. Hamming codes, on the other hand, are error-correcting codes capable of identifying and correcting single-bit errors using structured parity bits. Currently, most communication systems employ CRC for error detection and rely on retransmission mechanisms if an error is detected. Hamming codes are commonly used in memory systems (ECC RAM) and simple communication links where single-bit correction is sufficient. However, CRC cannot correct errors, and Hamming codes are limited in handling multi-bit or burst errors. Using either technique independently may not provide optimal reliability in environments with mixed error patterns. This project proposes the design and evaluation of an Adaptive Hybrid CRC–Hamming Error Control System that integrates Hamming coding (for correction) and CRC (for strong detection). The system will be analyzed under random and burst error conditions and evaluated using metrics such as detection rate, correction rate, undetected error probability, redundancy overhead, and effective throughput. The expected outcome is a reliability-enhanced error-control framework suitable for low-power communication systems, IoT devices, and embedded networks.

References

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